

SIMATS ENGINEERING

ASSESSMENT TOOL 2(MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1408

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.
2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.
3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.
4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.
5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities.

Answers

PORT SCANNING CHALLENGE — ENUMERATION ANALYSIS

1. Nmap Security Assessment – 192.168.1.20

Objective

To perform a security assessment of the server 192.168.1.20 using Nmap and identify open TCP ports, running services, service versions, and the exposed attack surface. The discovered services are analyzed based on their purpose and appropriate security measures are recommended.

Target

IP Address: 192.168.1.20

Tool: Nmap

Environment: Authorized laboratory/network environment

Procedure

First, verify whether the target host is active. Then perform a TCP port scan, identify running services and their versions, and use Nmap's default NSE scripts to obtain additional information.

A. Host Discovery

Command:

```
nmap -sn 192.168.1.20
```

Sample Output:

Starting Nmap 7.95

Nmap scan report for 192.168.1.20

Host is up (0.0021s latency).

Nmap done: 1 IP address (1 host up) scanned

Result:

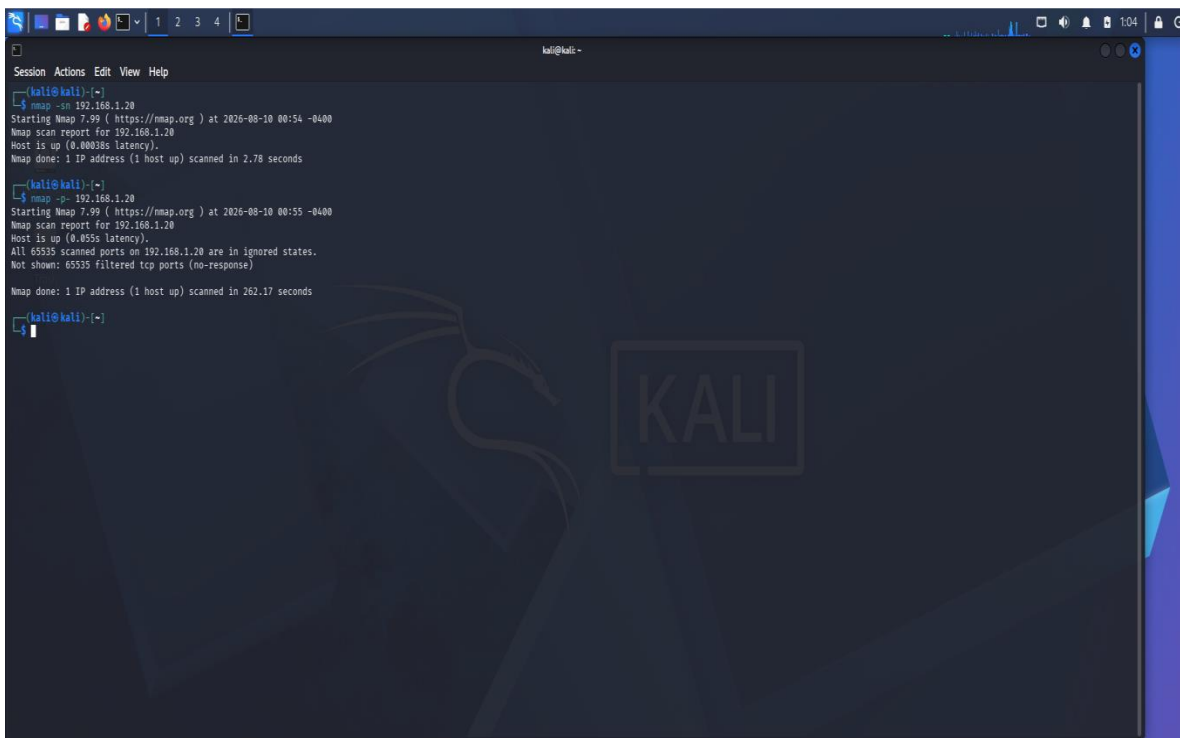
The target server 192.168.1.20 was detected as active.

B. TCP Port Scanning

To scan all TCP ports:

```
sudo nmap -sS -p- 192.168.1.20
```

Sample Output:



```
kali@kali:~$ nmap -sn 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:54 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00038s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.78 seconds

kali@kali:~$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:55 -0400
Nmap scan report for 192.168.1.20
Host is up (0.4055s latency).
All 65535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 262.17 seconds

kali@kali:~$
```

C. Service Version Detection

Command:

```
sudo nmap -sV 192.168.1.20
```

Sample Output:

```
kali@kali: ~  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:11 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.00056s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.70 seconds  
  
kali@kali: ~  
$ sudo nmap -sU --top-ports 100 192.168.1.20  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:13 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0022s latency).  
Not shown: 95 open/filtered udp ports (no-response)  
PORT      STATE SERVICE  
135/udp   filtered msrpc  
445/udp   filtered microsoft-ds  
1023/udp  filtered unknown  
5060/udp  filtered sip  
5353/udp  filtered zeroconf  
  
Nmap done: 1 IP address (1 host up) scanned in 37.14 seconds  
  
kali@kali: ~  
$ sudo nmap -Pn -SS -p 1-1000 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:15 -0400  
Stats: 0:13:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 17.80% done; ETC: 02:30 (1:02:02 remaining)  
  
kali@kali: ~  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:37 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.00051s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.57 seconds  
  
kali@kali: ~  
$
```

D. Default NSE Script Scanning

Command:

```
sudo nmap -sC -sV 192.168.1.20
```

**Sample
Result:**

```
kali@kali: ~  
$ sudo nmap -sU --top-ports 100 192.168.1.20  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:42 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0014s latency).  
Not shown: 96 open/filtered udp ports (no-response)  
PORT      STATE SERVICE  
17/udp    filtered qotd  
445/udp   filtered microsoft-ds  
1023/udp  filtered unknown  
4444/udp  filtered krb524  
  
Nmap done: 1 IP address (1 host up) scanned in 5.92 seconds  
  
kali@kali: ~  
$
```

E. Operating System Detection

Command:

```
sudo nmap -O 192.168.1.20
```

Security Analysis

The Nmap scan provides information about the network services exposed by 192.168.1.20. An attacker could use this information during reconnaissance to identify possible entry points.

- **SSH (22):** May be targeted for password guessing if weak authentication is permitted.
- **HTTP (80):** The web server may contain vulnerable applications, outdated components, or misconfigurations.
- **SMB/NetBIOS (139/445):** Improperly secured file-sharing services may expose system or shared-resource information.
- **MySQL (3306):** Direct database exposure increases the risk of unauthorized database access.

Service versions can also be compared with known security advisories to identify outdated components.

Recommendations

- Disable unnecessary services.
- Restrict SSH access to trusted administrative systems.
- Use key-based SSH authentication and strong passwords.
- Keep the web server and applications patched.
- Restrict SMB/NetBIOS to trusted internal networks.
- Do not expose MySQL directly to untrusted networks.
- Configure firewall rules using least privilege.
- Monitor repeated connection and authentication attempts.
- Perform periodic security scans after hardening.

Conclusion

Nmap successfully provides a detailed view of the target's exposed network services. Identifying open ports and service versions helps administrators reduce unnecessary exposure and prioritize security controls.

2. Nikto Web Server Assessment – 192.168.1.30

Objective

To assess the web server hosted on 192.168.1.30 using Nikto and identify potential web-server misconfigurations, outdated software components, exposed files, directories, and other security weaknesses.

Target

IP Address: 192.168.1.30

Tool: Nikto

A. Basic Web Server Scan

First check whether HTTP is accessible:

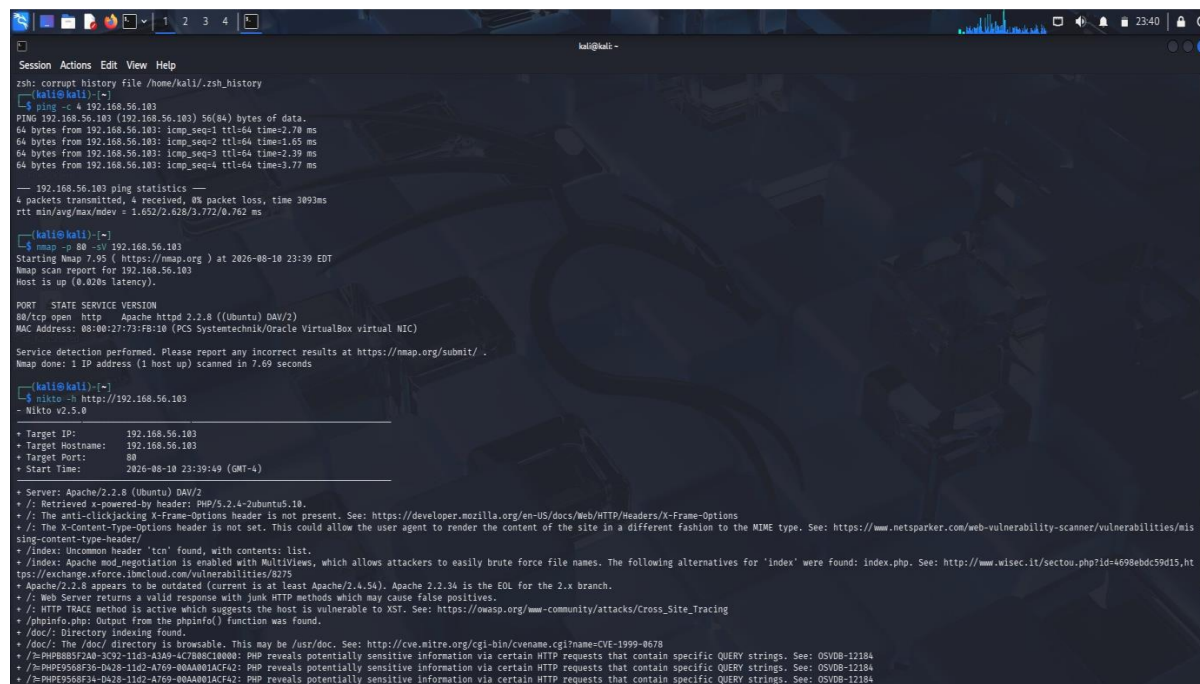
```
nmap -p 80,443 192.168.1.30
```

Sample Output:

```
No mail.
msfadmin@metasploitable:~$ sudo ifconfig eth0 192.168.56.103 netmask 255.255.255.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.103  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:519 errors:0 dropped:0 overruns:0 frame:0
          TX packets:77 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:34902 (34.0 KB)  TX bytes:10419 (10.1 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:118 errors:0 dropped:0 overruns:0 frame:0
          TX packets:118 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:31841 (31.0 KB)  TX bytes:31841 (31.0 KB)

msfadmin@metasploitable:~$
```



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.65 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=2.39 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=3.77 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3093ms
rtt min/avg/max/mdev = 1.652/2.628/3.772/8.762 ms

kali@kali:~$ nmap -p 80 -sV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2020-08-10 23:39 EDT
Nmap scan report for 192.168.56.103
Host is up (0.020s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:73:FB:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds

kali@kali:~$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2020-08-10 23:39:49 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/mis-sing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4698ebdc59d15,ht
+ /exchange.xforce.ibmcloud.com/vulnerabilities/6375
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross\_Site\_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0670
+ /h/PHP5588F36-D428-11D2-A769-00A0801ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /h/PHP5588F36-D428-11D2-A769-00A0801ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /h/PHP5588F36-D428-11D2-A769-00A0801ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
```

B. Nikto Scan

For HTTP:

```
nikto -h http://192.168.1.30
```

Sample Output:

```
Session Actions Edit View Help
80/tcp open  http    Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:73:FB:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds

kali@kali:~$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2026-08-10 23:39:49 (GMT+4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/mis-sing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4698ebdc59d15,ht-tps://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ /%PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /%PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /%PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/changelog: Server may leak inodes via ETags, header found with file /phpMyAdmin/changelog, inode: 92462, size: 48540, mtime: Tue Dec 9 12:24:00 2008. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ /phpMyAdmin/changelog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /test/: Directory indexing found.
+ /test/: This might be interesting.
+ /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ /icons/: Directory indexing found.
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ /phpMyAdmin/: phpMyAdmin directory found.
+ /phpMyAdmin/documentation.html: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/README: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts. See: https://typo3.org/
+ /#wp-config.php#: #wp-config.php# file found. This file contains the credentials.
+ 8910 requests: 0 error(s) and 27 item(s) reported on remote host
+ End Time: 2026-08-10 23:41:41 (GMT+4) (112 seconds)

+ 1 host(s) tested

kali@kali:~$
```

C. HTTPS Scan

If HTTPS is available:

```
nikto -h https://192.168.1.30
```

Findings

Finding	Possible Risk
Server version disclosed	Helps fingerprint the server
Missing security headers	May increase web attack exposure
Directory listing	May expose files and resources
Backup directory	Could expose sensitive information
/server-status	May disclose server information
Outdated components	May contain known vulnerabilities

Security Analysis

Nikto can identify common web-server configuration weaknesses and information disclosure. An attacker could use exposed directories and server-version information to understand the structure of the web application.

For example, a publicly accessible backup directory could potentially contain configuration files, source code, or old application data. Server-version disclosure can also help an attacker search for vulnerabilities associated with a specific software version.

Recommendations

- Remove unnecessary files and backup directories from the web root.
- Disable directory listing.

- Restrict administrative/status pages.
- Hide unnecessary server-version information.
- Configure security headers.
- Keep Apache/Nginx and application components updated.
- Use HTTPS with secure TLS configuration.
- Apply proper file and directory permissions.
- Regularly scan the web server for exposed resources.

Conclusion

Nikto provides useful information about web-server configuration and potentially exposed resources. Removing unnecessary files, applying security headers, restricting sensitive directories, and maintaining updated software can significantly reduce the web server's attack surface.

3. Netcat Investigation – 192.168.1.40

Objective

To investigate the host 192.168.1.40 using Netcat and collect available service banners from open ports. The collected banner information is analyzed to understand how attackers can use service and version information during reconnaissance.

Target

IP Address: 192.168.1.40

Tool: Netcat (nc)

Procedure

First identify possible open ports using Nmap:

```
nmap -sV 192.168.1.40
```

Sample Output:

A. Connect to SSH

```
nc -v 192.168.1.40 22
```

B. Connect to HTTP

```
nc -v 192.168.1.40 80
```

Then type:

```
HEAD / HTTP/1.0
```

Press **Enter twice**.

Sample Output:

C. SMTP Banner

```
nc -v 192.168.1.40 25
```


Banner Information Analysis

The banners may reveal:

- Service name
- Software name
- Software version
- Operating system information
- Server type
- Protocol information

For example:

OpenSSH_8.2

Apache/2.4.41

Postfix

can provide an attacker with useful information about the technologies deployed on the server.

How Attackers Could Use Banner Information

An attacker performing reconnaissance could:

1. Identify the running service.
2. Determine the software version.
3. Search vulnerability databases for that version.
4. Identify possible weaknesses.
5. Select further enumeration techniques.
6. Prioritize services for security testing.

Therefore, banner information contributes to **fingerprinting** the target system.

Security Recommendations

- Minimize unnecessary service-banner information.
- Keep services updated.
- Disable unnecessary services.
- Restrict access using firewalls.
- Use secure authentication mechanisms.
- Monitor suspicious connection attempts.
- Avoid exposing internal hostnames unnecessarily.

Conclusion

Netcat allows a security analyst to manually connect to services and observe available banners. Banner information can significantly assist reconnaissance, so unnecessary version and system information should be minimized.

4. Telnet Examination – 192.168.1.50

Objective

To examine the server 192.168.1.50 using Telnet and determine whether remote administration or other TCP services are accessible. The security risks of insecure remote administration protocols are analyzed and secure alternatives are recommended.

Target

IP Address: 192.168.1.50

Tool: Telnet

Procedure

First identify possible open ports:

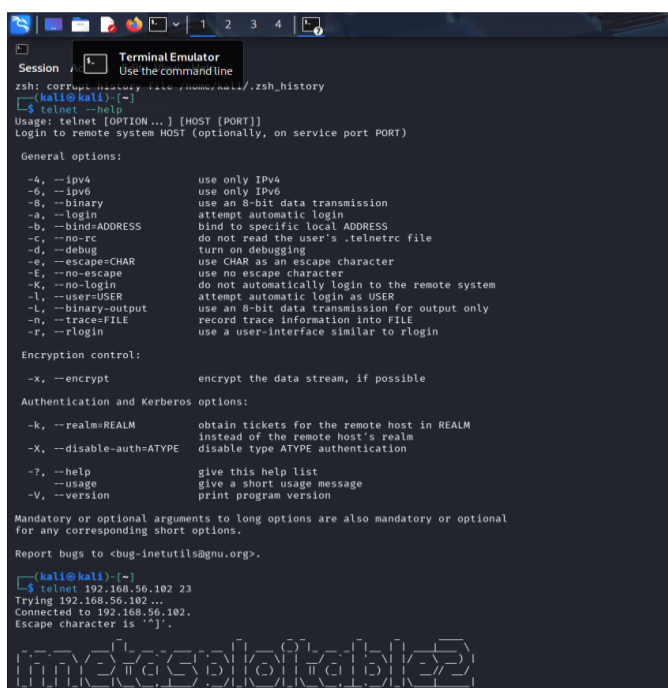
```
nmap -sV 192.168.1.50
```

A. Check Telnet Port

```
telnet 192.168.1.50 23
```

B. Check SSH

```
telnet 192.168.1.50 22
```



```
zsh: COFFUP history file /home/kali/.zsh_history
(kali@kali)~$ telnet --help
Usage: telnet [OPTION...] [HOST [PORT]]
Login to remote system HOST (optionally, on service port PORT)

General options:
  -4, --ipv4           use only IPv4
  -6, --ipv6           use only IPv6
  -8, --binary         use an 8-bit data transmission
  -a, --login          attempt automatic login
  -b, --bind=ADDRESS   bind to specific local ADDRESS
  -c, --no-rc          do not read the user's .telnetrc file
  -d, --debug          turn on debugging
  -e, --escape=CHAR    use CHAR as an escape character
  -E, --no-escape      use no escape character
  -k, --no-login       do not automatically login to the remote system
  -l, --user=USER      attempt automatic login as USER
  -L, --binary-output  use an 8-bit data transmission for output only
  -n, --trace=FILE    record trace information into FILE
  -r, --rlogin         use a user-interface similar to rlogin

Encryption control:
  -x, --encrypt        encrypt the data stream, if possible

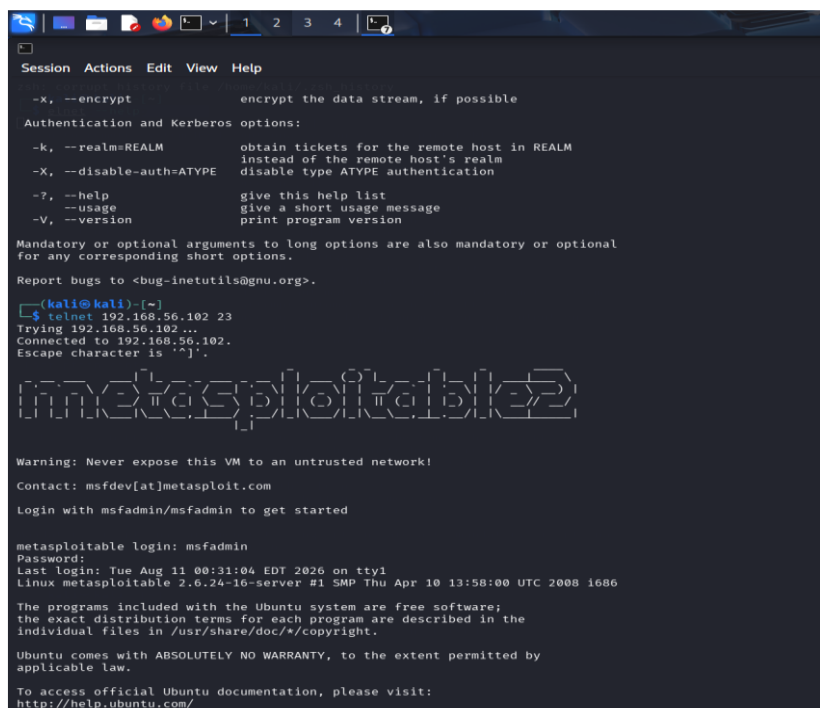
Authentication and Kerberos options:
  -k, --realm=REALM    obtain tickets for the remote host in REALM
                        instead of the remote host's realm
  -X, --disable-auth=ATYPE disable type ATYPE authentication
  -?, --help           give this help list
  --usage             give a short usage message
  -V, --version        print program version

Mandatory or optional arguments to long options are also mandatory or optional
for any corresponding short options.

Report bugs to <bug-inetutils@gnu.org>.

(kali@kali)~$ telnet 192.168.56.102 23
Trying 192.168.56.102 ...
Connected to 192.168.56.102.
Escape character is '^]'.

metasploitable2
```



```
Session Actions Edit View Help

-x, --encrypt           encrypt the data stream, if possible

Authentication and Kerberos options:
-k, --realm=REALM       obtain tickets for the remote host in REALM
                        instead of the remote host's realm
-X, --disable-auth=ATYPE disable type ATYPE authentication
-?, --help             give this help list
--usage               give a short usage message
-V, --version          print program version

Mandatory or optional arguments to long options are also mandatory or optional
for any corresponding short options.

Report bugs to <bug-inetutils@gnu.org>.

(kali@kali)~$ telnet 192.168.56.102 23
Trying 192.168.56.102 ...
Connected to 192.168.56.102.
Escape character is '^]'.

metasploitable2

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com

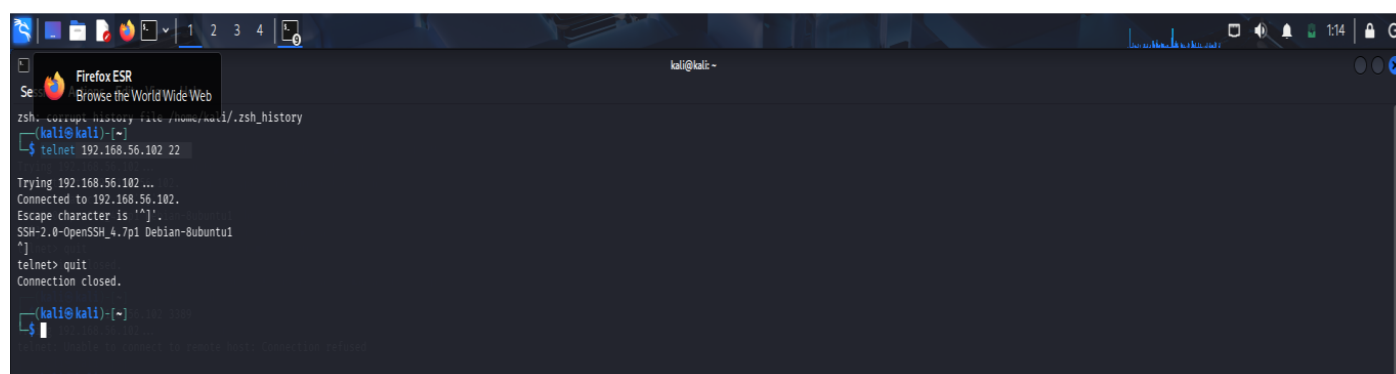
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Tue Aug 11 00:31:04 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
```



```
zsh: COFFUP history file /home/kali/.zsh_history
(kali@kali)~$ telnet 192.168.56.102 22
Trying 192.168.56.102 ...
Connected to 192.168.56.102.
Escape character is '^]'.
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
telnet> quit
Connection closed.

(kali@kali)~$
```

Security Analysis

Telnet is insecure for remote administration because traditional Telnet communication does not provide strong encryption. Credentials and other transmitted information can potentially be intercepted by an attacker who can observe the network traffic.

An exposed Telnet service therefore creates a significant security risk.

An attacker could use the discovered service to:

- Identify remote administration capability.
- Attempt unauthorized authentication.
- Capture unencrypted credentials in an appropriate network position.
- Fingerprint the remote system.
- Perform further service enumeration.

Recommendations

- Disable Telnet completely if it is not required.
- Replace Telnet with **SSH**.
- Restrict SSH to authorized management networks.
- Use strong authentication.
- Prefer SSH key-based authentication.
- Disable direct root login where appropriate.
- Apply firewall restrictions.
- Monitor failed login attempts.
- Keep remote administration software updated.

Secure Alternative

Instead of:

Telnet → Port 23

use:

SSH → Port 22

SSH provides encrypted remote administration and is substantially safer than Telnet.

Conclusion

Telnet examination can identify whether insecure remote administration services are exposed. If Telnet is discovered on 192.168.1.50, it should normally be disabled and replaced with securely configured SSH.

5. Netdiscover Host Discovery – Network 192.168.1.1–50

Objective

To perform host discovery on the local network segment 192.168.1.1–50 using Netdiscover and identify active hosts. The discovered IP and MAC address information is analyzed from a security perspective.

Target Network

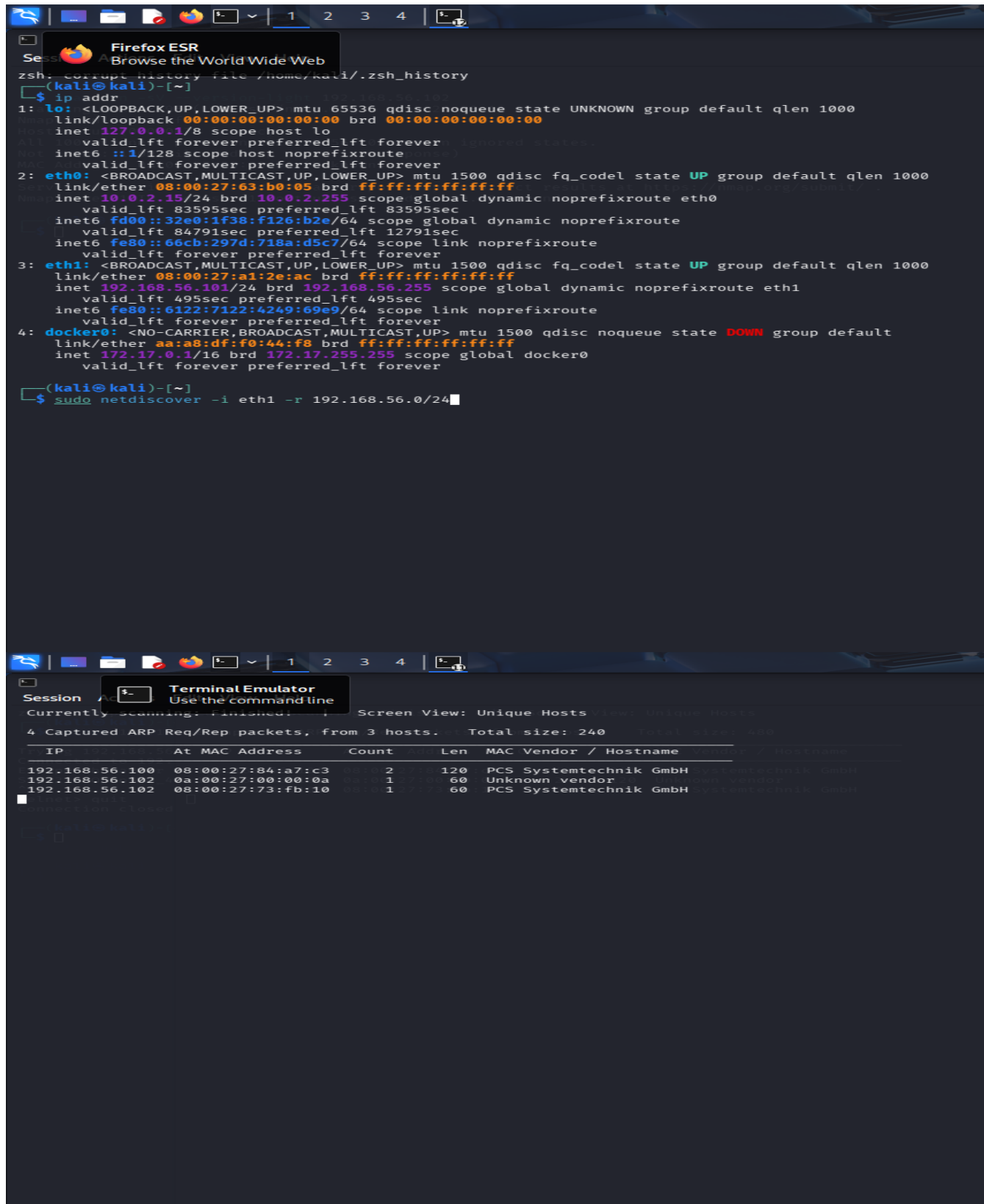
Network Range: 192.168.1.1–192.168.1.50

Tool: Netdiscover

A. Netdiscover Scan

`sudo netdiscover -r 192.168.1.0/24`

This scans the local /24 network.



Findings

The host-discovery process can provide:

- Active IP addresses
- MAC addresses
- Hardware/vendor information
- Number of responding systems
- Potential network infrastructure devices

Security Analysis

Host discovery is an important reconnaissance step because it allows an attacker to build a basic map of the network.

For example:

192.168.1.1 → Gateway

192.168.1.20 → Server

192.168.1.30 → Web server

192.168.1.40 → Service host

192.168.1.50 → Administration server

Once active hosts are identified, an attacker could perform further enumeration against those systems.

The attacker may then use tools such as Nmap to identify:

- Open ports
- Running services
- Service versions
- Operating systems
- Network protocols

This demonstrates why unnecessary network exposure should be minimized.

Recommendations

- Use network segmentation.
- Apply firewall rules between network zones.
- Restrict access to administrative systems.
- Disable unused network services.
- Monitor unusual ARP/network-discovery activity.
- Maintain an inventory of authorized devices.
- Use NAC where appropriate.
- Isolate critical servers from ordinary user networks.
- Regularly review network exposure.

Conclusion

Netdiscover successfully identifies active hosts on a local network by observing ARP-based communication. The resulting IP, MAC, and vendor information can assist legitimate administrators with network inventory but can also provide attackers with a foundation for further enumeration. Proper segmentation, firewall controls, monitoring, and device management can reduce this reconnaissance exposure.

Overall Security Assessment

Question	Tool	Target	Main Purpose
1	Nmap	192.168.1.20	Port and service enumeration
2	Nikto	192.168.1.30	Web-server security assessment
3	Netcat	192.168.1.40	Service banner collection
4	Telnet	192.168.1.50	Remote-service accessibility check
5	Netdiscover	192.168.1.1–50	Network host discovery

Code of Conduct:

*I ____ C Prathibha (192424293)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C Prathibha

Q.No	Identification of Port Scanning Factors(25)	Analysis(25)	Linkage(20)	Application(20)	Clarity(10)	Marks
1						
2						
3						
4						
5						

Signature of Faculty

Total Avg :

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight
Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand

